EXPERIMENT 14 : Complex number addition using class

A] ALGORITHM;

Step 1: Start

Step 2: Define a class Complex with attributes:

- real
- imag

Step 3: Define method initComplex()

- Read two integers from user input
- Assign them to real and imag

Step 4: Define method sum(c1, c2)

- Set real = c1.real + c2.real
- Set imag = c1.imag + c2.imag

Step 5: Define method display()

- If imag >= 0, print "real + imag i"
- Else, print "real - |imag| i"

Step 6: Create three objects:

- c1, c2, c3 of class Complex

Step 7: Call initComplex() for c1

- Input first complex number

Step 8: Call initComplex() for c2

- Input second complex number

Step 9: Call sum(c1, c2) using c3

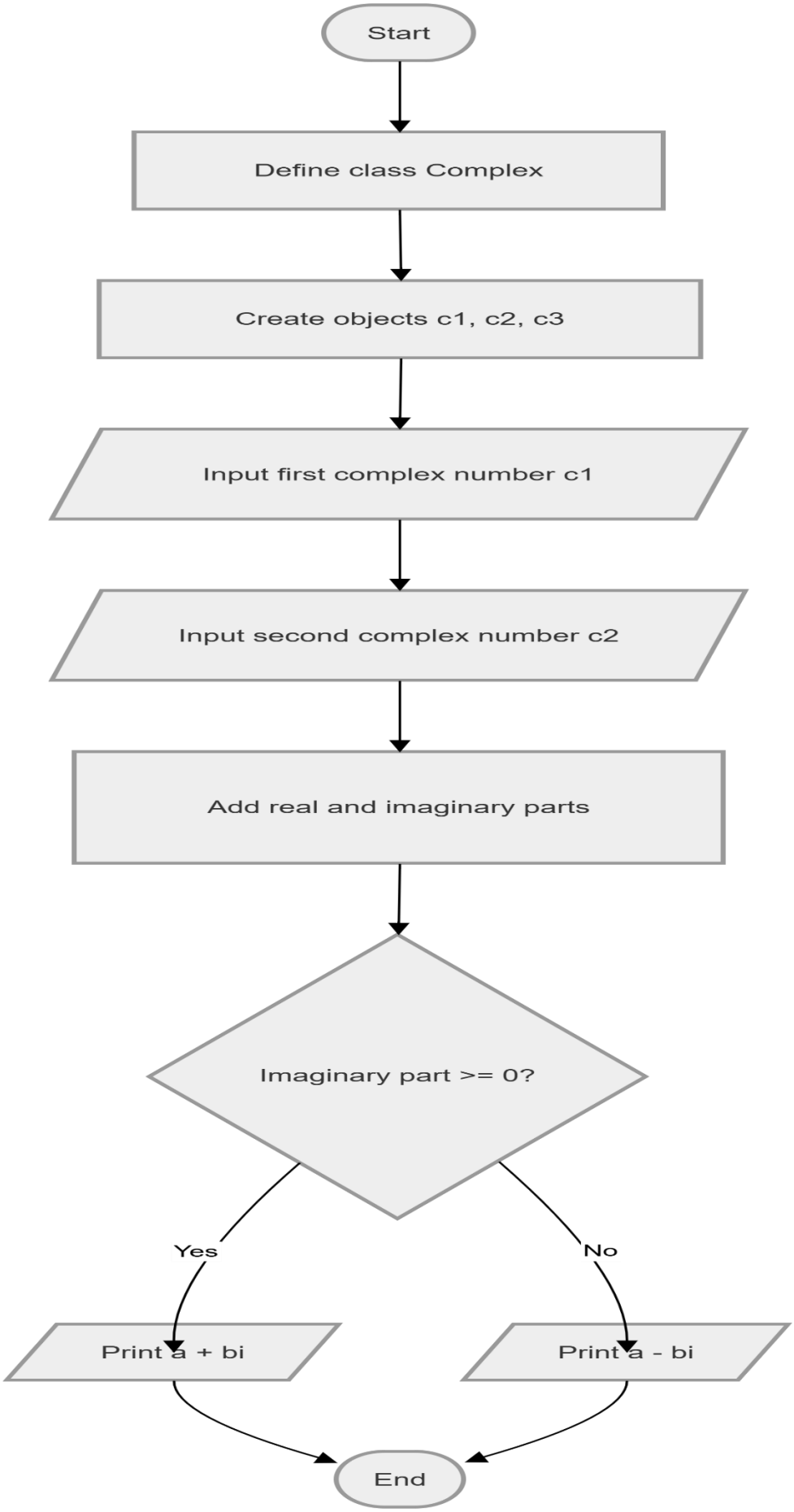
- Store result in c3

Step 10: Call display() using c3

- Display the result

Step 11: Stop

B]FLOWCHART:


Start
Define class Complex
Create objects c1, c2, c3
Input first complex number c1
Input second complex number c2
Add real and imaginary parts
Imaginary part >= 0?
Yes
No
Print a + bi
Print a - bi
End

C] code:

```
class Complex:
    def initComplex(self):
        self.real,self.imag = map(int,input().split())

    def sum(self,C1,C2):
        self.real = C1.real + C2.real
        self.imag = C1.imag + C2.imag

    def display(self):
        if self.imag>= 0:
            print(f"{self.real} + {self.imag}i")
        else:
            print(f"{self.real} - {abs(self.imag)}i")



# Create three instances
c1 = Complex()
c2 = Complex()
c3 = Complex()

# Initialize two complex numbers
c1.initComplex()
c2.initComplex()

# Compute and display sum
c3.sum(c1, c2)
c3.display()
```

D]CODETANTRA:

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14.1.1. Complex Number Addition using Class 22:36

Write a Python program using a class named **Complex** to add two complex numbers.

Create a class **Complex** with the following methods:

- **initComplex()**: Read user input for the real and imaginary parts.
- **sum()**: Compute the sum of two complex numbers and store the result in the current instance.
- **display()**: Display the complex number in the format "<a> + <b>i" or "<a> - <b>i".

The program should create three instances of **Complex** class: two for input complex numbers and one for the result. Initialize both complex numbers, compute their sum, and display the result.

Input Format:

- First line contains two space-separated numbers: real and imaginary parts of first complex number
- Second line contains two space-separated numbers: real and imaginary parts of second complex number

Output Format:

- Print a single line showing the sum of the two complex numbers in the format "<a> + <b>i" or "<a> - <b>i".

Sample Test Cases +

Explorer

complex... Submit

```
class Complex:
    def initComplex(self):
        self.real,self.imag = map(int,input().split())

    def sum(self,C1,C2):
        self.real = C1.real + C2.real
        self.imag = C1.imag + C2.imag

    def display(self):
        if self.imag>= 0:
            print(f"{self.real} + {self.imag}i")
        else:
            print(f"{self.real} - {abs(self.imag)}i")
```

Debugger

Average time **0.014 s** 13.75 ms

Maximum time **0.018 s** 18.00 ms

4 out of 4 shown test case(s) passed

4 out of 4 hidden test case(s) passed

Test case 1 18 ms Debug

Expected output	Actual output
3 4	3 4
2 5	2 5
5 + 9i	5 + 9i

Test case 2 13 ms

Terminal Test cases